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DETERMINATION OF QUANTUM CIRCUIT COMPILATION PASSES AND/OR COMPILATION PARAMETERS OPTIMIZED FOR ONE OR MORE PERFORMANCE METRICS

Abstract

A quantum computing service includes a quantum circuit compilation module that uses received information about quantum processing units (QPUs) along with information about a quantum circuit to be compiled in order to generate an optimized ordered list of compilation passes to be performed to compile the quantum circuit for execution on a given QPU.

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Background/Summary

PRIORITY CLAIM [0001] This application is a continuation of U.S. patent application Ser. No. 17/491,180, filed Sep. 30, 2021, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] Quantum computing utilizes the laws of quantum physics to process information. Quantum physics is a theory that describes the behavior of reality at the fundamental level. It is currently the only physical theory that is capable of consistently predicting the behavior of microscopic quantum objects like photons, molecules, atoms, and electrons.

[0003] A quantum computer is a device that utilizes quantum physics to allow one to write, store, process and read out information encoded in quantum states, e.g., the states of quantum objects. A quantum object is a physical object that behaves according to the laws of quantum physics. The state of a physical object is a description of the object at a given time.

[0004] In quantum physics, the state of a two-level quantum system, or simply, a qubit, is a list of two complex numbers whose squares sum up to one. Each of the two numbers is called an amplitude, or quasi-probability, and their squared absolute values are probabilities that a measurement of the qubit results in zero or one. A fundamental and counterintuitive difference between a probabilistic bit (e.g., a classical zero or one bit) and the qubit is that a probabilistic bit represents a lack of information about a two-level classical system, while a qubit contains maximal information about a two-level quantum system.

[0005] Quantum computers are based on such quantum bits (qubits), which may experience the phenomena of “superposition” and “entanglement.” Superposition allows a quantum system to be in multiple states at the same time. For example, whereas a classical computer is based on bits that are either zero or one, a qubit may be both zero and one at the same time, with different probabilities assigned to zero and one. Entanglement is a strong correlation between quantum systems, such that the quantum systems are inextricably linked even if separated by great distances.

[0006] A quantum algorithm comprises a reversible transformation acting on qubits in a desired and controlled way, followed by a measurement on one or multiple qubits. For example, if a system has two qubits, a transformation may modify four numbers; with three qubits this becomes eight numbers, and so on. As such, a quantum algorithm acts on a list of numbers exponentially large as dictated by the number of qubits. To implement a transform, the transform may be decomposed into small operations acting on a single qubit, or a pair of qubits, as an example. Such small operations may be called quantum gates and a specific arrangement of the quantum gates implements a quantum circuit.

[0007] There are different types of qubits that may be used in quantum computers, each having different advantages and disadvantages. For example, some quantum computers may include qubits built from superconductors, trapped ions, semiconductors, photonics, etc. Each may experience different levels of interference, errors and decoherence. Also, some may be more useful for generating particular types of quantum circuits or quantum algorithms, while others may be more useful for generating other types of quantum circuits or quantum algorithms. Also, costs, run-times, error rates, availability, etc. may vary across quantum computing technologies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates a quantum computing service of a service provider network that enables customers to access quantum computers that use multiple quantum computing technologies, wherein the quantum computing service includes a quantum circuit compilation pass and/or

compilation parameters determination module, according to some embodiments.

[0009] FIG. 2A illustrates a more detailed view of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0010] FIG. 2B illustrates a more detailed view of an example compilation optimization module of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0011] FIG. 2C illustrates a more detailed view of another example compilation optimization module of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0012] FIG. 2D illustrates a more detailed view of yet another example compilation optimization module of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0013] FIG. 3 illustrates an example directed acyclic graph that may be used to represent a quantum circuit, according to some embodiments.

[0014] FIG. 4 illustrates example compilation passes that may be supported by a compilation module of a quantum computing service and from which compilation passes may be selected for inclusion in an optimized list of quantum circuit compilation passes (and/or compilation parameters) to be used to compile a given quantum circuit, according to some embodiments.

[0015] FIG. 5 illustrates example quantum processing unit (QPU) information that may be used by a compilation module of a quantum computing service to determine an optimized list of quantum circuit compilation passes (and/or compilation parameters) to be used to compile a given quantum circuit, according to some embodiments.

[0016] FIG. 6 illustrates example quantum circuit description information that may be used by a compilation module of a quantum computing service to determine an optimized list of quantum circuit compilation passes (and/or compilation parameters) to be used to compile a given quantum circuit, according to some embodiments.

[0017] FIG. 7 illustrates example compiler constraint information that may be used by a compilation module of a quantum computing service to determine an optimized list of quantum circuit compilation passes (and/or compilation parameters) to be used to compile a given quantum circuit, according to some embodiments.

[0018] FIG. 8 illustrates an example process that may be followed by a compilation module of a quantum computing service to determine an optimized list of quantum circuit compilation passes (and/or compilation parameters) to be used to compile a given quantum circuit, according to some embodiments.

[0019] FIG. 9 illustrates an example compilation optimization process that may be used by a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0020] FIG. 10 illustrates another example compilation optimization process that may be used by a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0021] FIG. 11 illustrates yet another example compilation optimization process that may be used by a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0022] FIG. 12 illustrates edge computing devices of a quantum computing service physically located at quantum hardware provider locations, according to some embodiments.

[0023] FIG. 13 illustrates an example edge computing device connected to a quantum computing service, according to some embodiments.

[0024] FIG. 14 is a block diagram illustrating an example classical computing device that may be used in at least some embodiments.

[0025] While embodiments are described herein by way of example for several embodiments and

illustrative drawings, those skilled in the art will recognize that embodiments are not limited to the embodiments or drawings described. It should be understood, that the drawings and detailed description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to. When used in the claims, the term “or” is used as an inclusive or and not as an exclusive or. For example, the phrase “at least one of x, y, or z” means any one of x, y, and z, as well as any combination thereof.

DETAILED DESCRIPTION

[0026] The present disclosure relates to methods and apparatus for providing quantum computing services to customers and enabling customers to seamlessly use one or more quantum computing technologies. Additionally, the disclosure relates to a compilation module of a quantum computing service that determines a list of compilation passes and/or compilation parameters to be used to compile a given quantum circuit for execution on a quantum processing unit (QPU). The determined list of compilation passes and/or compilation parameters may be determined in a way that optimizes one or more optimization metrics (M) for the quantum circuit, such as a run-time for a compiled version of the quantum circuit, a complexity of a compiled version of the quantum circuit, an error-rate for a compiled version of the quantum circuit, and/or various other optimization criteria. In some embodiments, a customer of the quantum computing service may select which optimization parameters are to be given priority when determining compilation passes and/or compilation parameters for quantum programs or quantum circuits submitted to the quantum computing service on behalf of the customer.

[0027] In some embodiments, a system includes a service provider network comprising one or more computing devices that are configured to implement a quantum computing service. The system may also include a first edge computing device of the service provider network located at a location of a first quantum hardware provider and one or more additional edge computing devices of the service provider network located at respective locations of one or more additional quantum hardware providers. In some embodiments, the quantum computing service may additionally, or alternatively, be connected to one or more quantum hardware devices located within the service provider network. For example, the quantum computing service may provide a customer access to internal quantum processing units (QPUs) and/or external quantum processing units of external quantum hardware providers (QPUs of QHPs). The one or more computing devices that implement the quantum computing service are configured to receive, from a customer of the quantum computing service, a definition of a quantum computing object to be executed and select at least one of the quantum hardware providers to execute the quantum computing object, or select an internal QPU to execute the quantum computing object. In some embodiments, the quantum computing object may be a quantum task or a quantum job, such as a task or job defined using a problem-domain interface of a quantum algorithm development kit; a quantum program defined using a quantum algorithm development kit of the quantum computing service or provided by a customer; or a quantum circuit defined using the quantum algorithm development kit of the quantum computing service, or supplied by a customer. In some embodiments, a quantum job may comprise a plurality of quantum tasks, such as quantum circuits to be executed. Also, a quantum program may include logical loops that instruct a particular quantum circuit to be implemented based on one or more conditions, such as a “for loop”, “if loop”, “while loop”, etc. In addition, in some embodiments, a quantum program may instruct implementation of a parametric quantum circuit. In such examples as described above, execution of an individual quantum circuit of a quantum program may be referred to herein as a quantum task and execution of the quantum

program as a whole may be referred to herein as a quantum job.

[0028] In some embodiments, a list of compilation passes and/or compilation parameters that may be used to compile a quantum program and/or quantum circuit submitted to the quantum computing service may be determined based on received information about the QPU that is to be used to execute the quantum circuit or a given quantum circuit of a quantum program. Additionally, the list of compilation passes and/or the compilation parameters may be determined in a way that optimizes one or more metrics (M) for the quantum circuit or the quantum program, such as a time required to execute the quantum circuit or quantum program (e.g., run-time), a complexity of the quantum circuit (e.g. number of physical qubits and/or gates required to implement the compiled quantum circuit) or a complexity of an overall quantum program comprising multiple quantum circuits, an error-rate for the compiled quantum circuit or quantum program, etc. In some embodiments, one or more default optimization metrics (M) may be used in an optimization algorithm to determine the compilation passes and/or the compilation parameters. In some embodiments, a customer that submits a quantum circuit to a quantum computing service for execution may additionally indicate one or more optimization metrics (M) to be used to optimize compilation of the quantum circuit.

[0029] In some embodiments, a compilation module of a quantum computing service may provide a determined list of compilation passes and/or compilation parameters to a quantum hardware provider (QHP) for use in compiling a quantum circuit at the QHP for execution on a quantum processing unit (QPU) of the QHP. Also, in some embodiments, the quantum computing service may further include a compiler, such as in the compilation module, that is configured to use the determined list of compilation passes and/or the compilation parameters to generate a QPU specific compiled binary, wherein the QPU specific compiled binary is sent to a QHP for execution on a QPU of the QHP. Also, in some embodiments, a quantum computing service may include internal quantum hardware devices that include QPUs and the compiled binary may be used to locally execute the quantum circuit on a QPU of the quantum computing service.

[0030] In some embodiments, a machine learning algorithm used by a compilation module of a quantum computing service may be trained using previously compiled quantum circuits and execution performance information for the previously compiled quantum circuits in relation to the optimization metrics (M). The trained machine learning algorithm may further be used to determine optimized lists of ordered compilation passes and/or compilation parameters to be used to compile a given quantum circuit in a way that is optimized for the optimization metric (M). In some embodiments, reinforcement learning may be used to train the machine learning model. In some embodiments, a tabular learning approach may be used.

[0031] In some embodiments, the one or more computing devices that implement the quantum computing service may further be configured to provide a customer with a recommendation in regard to which quantum hardware provider or internal QPU to use to execute the customer's quantum computing object (for example to better achieve the customer's specified optimization metric (M)). In some embodiments, selection of an internal QPU of the quantum hardware provider or an external QPU of a QHP to execute the quantum computing object may be based on the recommendation generated by the quantum computing service and/or other input received from the customer regarding which type of quantum hardware to select to execute the customer's quantum computing object. In some embodiments, the recommendation may include estimated costs, error rates, run-times, etc. associated with executing the quantum computing object on quantum computers of respective ones of the quantum hardware providers or an internal QPU. In some embodiments, optimized ordered compilation passes and/or compilation parameters may be determined for a specific QPU, once a customer has selected which QHP and/or QPU is to be used to execute the customer's quantum circuit. Also, in some embodiments, optimized ordered compilation passes and/or compilation parameters may be determined for a simulation, such as simulation that simulates execution of a quantum circuit on a particular type of quantum computing

device and/or a particular QPU.

[0032] The one or more computing devices implementing the quantum computing service are further configured to submit a quantum task comprising a quantum circuit to a selected one or more of the quantum hardware providers via a given edge computing device located at the respective location of the selected quantum hardware provider, receive results of executing the quantum circuit on a quantum computer of the selected quantum hardware provider, store the results of executing the quantum circuit, and provide a notification to the customer that execution of the quantum computing object has been completed. However, when an internal QPU is selected for execution of the quantum task, the quantum task may be submitted directly to a queue for the internal QPU. For example, it may not be necessary to transport the quantum task, via back-end API transport, to an edge computing device for submission of a quantum task to an internal QPU.

[0033] Note that in some embodiments, a quantum task comprising a quantum circuit may be sent to a QHP or internal QPU as a compiled binary that has been compiled using an optimized ordered list of compilation passes and/or compilation parameters as determined by the compilation module of the quantum computing service. In some embodiments, the quantum task comprising the quantum circuit may be sent to a QHP as a quantum circuit definition and accompanying ordered list of compilation passes and/or compilation parameters to be used by a compiler of the QHP to generate a compiled binary for executing the quantum circuit on a QPU of the QHP.

[0034] In some embodiments, the quantum computing service is also configured to translate a quantum computing object into a selected quantum circuit format for a particular quantum computing technology used by the selected quantum hardware provider or internal QPU, wherein the selected quantum circuit format for the particular quantum computing technology is one of a plurality of quantum circuit formats for a plurality of different quantum computing technologies supported by the quantum computing service. In some embodiments, the quantum circuit may be translated into a quantum hardware specific quantum circuit definition format prior to being provided to a compilation module in order to determine an optimized ordered list of compilation passes and/or compilation parameters for the quantum circuit.

[0035] To translate the quantum computing object into the selected quantum circuit format, the one or more computing devices that implement the quantum computing service are configured to identify portions of the quantum computing object corresponding to quantum operators in an intermediate representation in which the quantum object was submitted by the customer, substitute the quantum operators of the intermediate representation with quantum operators of the quantum circuit format of the particular quantum computing technology, and perform one or more optimizations to reduce an overall number of quantum operators in a translated quantum circuit that is a translated version of the received quantum computing object. In some embodiments, translation may be omitted or combined with compilation. Also, in some embodiments, some aspects of translation may be performed by a compiler, wherein a compilation module determines optimized list of passes that may include both translation passes and compilation passes.

Example Quantum Computing Service

[0036] Quantum computers may be difficult and costly to construct and operate. Also, there are varying quantum computing technologies under development with no clear trend as to which of the developing quantum computing technologies may gain prominence. Thus, potential users of quantum computers may be hesitant to invest in building or acquiring a particular type of quantum computer, as other quantum computing technologies may eclipse a selected quantum computing technology that a potential quantum computer user may invest in. Also, successfully using quantum computers to solve practical problems may require significant trial and error and/or otherwise require significant expertise in using quantum computers.

[0037] As an alternative to building and maintaining a quantum computer, potential users of quantum computers may instead prefer to rely on a quantum computing service to provide access to quantum computers. Also, in some embodiments, a quantum computing service, as described

herein, may enable potential users of quantum computers to access quantum computers based on multiple different quantum computing technologies and/or paradigms, without the cost and resources required to build or manage such quantum computers. Also, in some embodiments, a quantum computing service, as described herein, may provide various services that simplify the experience of using a quantum computer such that potential quantum computer users lacking deep experience or knowledge of quantum mechanics, may, never the less, utilize quantum computing services to solve problems.

[0038] Also, in some embodiments, a quantum computing service, as described herein, may be used to supplement other services offered by a service provider network. For example, a quantum computing service may interact with a classical computing service to execute hybrid algorithms. In some embodiments, a quantum computing service may allow a classical computer to be accelerated by sending particular tasks to a quantum computer for execution, and then further performing additional classical compute operations using the results of the execution of a quantum computing object on the quantum computer. For example, a quantum computing service may allow for the acceleration of virtual machines implemented on classical hardware in a similar manner as a graphics processing unit (GPU) may accelerate graphical operations that otherwise would be performed on a central processing unit (CPU).

[0039] In some embodiments, a quantum computing service may provide potential quantum computer users with access to quantum computers using various quantum computing technologies, such as quantum annealers, ion trap machines, superconducting machines, photonic devices, etc. In some embodiments, a quantum computing service may provide customers with access to at least three broad categories of quantum computers including quantum annealers, circuit-based quantum computers, and analog or continuous variable quantum computers. As used herein, these three broad categories may be referred to as quantum computing paradigms.

[0040] In some embodiments, a quantum computing service may be configured to provide simulation services using classical hardware based computing instances to simulate execution of a quantum circuit on a quantum computer. In some embodiments, a quantum computing service may be configured to perform general simulation and/or simulation that specifically simulates execution of a quantum circuit on a particular type of quantum computer of a particular quantum computer technology type or paradigm type. In some embodiments, simulation may be fully managed by a quantum computing service on behalf of a customer of the quantum computing service. For example, the quantum computing service may reserve sufficient computing capacity on a virtualized computing service of the service provider network to perform simulation without customer involvement in the details of managing the resources for the simulator.

[0041] In some embodiments, a quantum computing service may include a dedicated console that provides customers access to multiple quantum computing technologies. Furthermore, the quantum computing service may provide a quantum algorithm development kit that enables customers with varying levels of familiarity with quantum circuit design to design and execute quantum circuits. In some embodiments, a console of a quantum computing service may include various application programmatic interfaces (APIs), such as: [0042] (Create/Delete/Update/Get/List) Simulator-Configuration-create, read, update, and delete (CRUD) operations for simulator configuration objects. [0043] (Start/Cancel/Describe) Simulator-used to control each of the user-defined simulator instances. [0044] (List/Describe) quantum processor units (QPU)s—retrieves quantum computer hardware information. [0045] (Create/Cancel/List/Describe) Job-used to manage the lifecycle of a quantum job. [0046] (Assign/Update/List) Quality of Service (QoS) guarantee-used to manage QoS guarantees for quantum jobs and/or quantum tasks. [0047] (Create/Cancel/List/Describe) Task-used to manage the lifecycle of individual quantum tasks/quantum objects.

[0048] In some embodiments, a quantum algorithm development kit may include a graphical user interface, APIs or other interface to allow customers of a quantum computing service to define

quantum objects, such as quantum tasks, algorithms or circuits, using the quantum algorithm development kit. In the some embodiments, the quantum algorithm development kit may include an interface option that enables customers to share the quantum objects with other customers of the quantum computing service. For example, the quantum algorithm development kit may include a marketplace that allows customers to share or sell particular quantum objects with other customers. In some embodiments, the quantum algorithm development kit may include an interface element that allows customers to select a QoS to be applied for a quantum job or quantum tasks defined via the quantum algorithm development kit.

[0049] In some embodiments, a quantum computing service may include a public application programmatic interface (API) that accepts quantum objects submitted by a customer of the quantum computing service. Additionally, the quantum computing service may include a back-end API transport that is non-public. The back-end API transport may enable quantum circuits to be transported from a centralized location that implements the quantum computing service, such as one or more data centers of a service provider network, to an edge computing device at a particular quantum hardware provider location where the quantum circuit is to be executed. In some embodiments, quantum objects or quantum tasks may be executed using an internal QPU of the quantum computing service without using a back-end API transport to transport the quantum job or quantum task to an external quantum hardware provider location.

[0050] In some embodiments, results of the execution of a quantum circuit on a quantum computer at a quantum hardware provider location may be provided to the edge computing device at the quantum hardware provider location. The edge computing device may automatically transport the results to a secure storage service of the service provider network, where the customer can access the results using the storage service of the service provider network or via a console of the quantum computing service. Likewise, results of execution of a quantum circuit via an internal QPU may be accessed via the console of the quantum computing service.

[0051] In some embodiments, the results stored to the secure storage service may be seamlessly used by other services integrated into the service provider network, such as a machine learning service, a database service, an object-based storage service, a block-storage service, a data presentation service (that reformats the results into a more usable configuration), etc. For example, in some embodiments, a machine learning service may be used to optimize a quantum algorithm or quantum circuit. For example, the machine learning service may cause various versions of a quantum algorithm or quantum circuit to be run on a quantum computer via a quantum computing service. The machine learning service may also be provided access to results of running the quantum algorithms or quantum circuits. In some embodiments, the machine learning service may cause the quantum algorithms or quantum circuits to be run on various different quantum computing technology based quantum computers. Based on the results, the machine learning service may determine one or more optimizations to improve the quantum algorithms or quantum circuits.

[0052] In some embodiments, a quantum computing service may support creating snapshots of results of executing a quantum circuit. For example, the quantum computing service may store snapshots of intermediate results of a hybrid algorithm or may more generally store snapshots of any results generated by executing a quantum circuit on a quantum computer. In some embodiments, an edge computing device at a hardware provider location may temporarily store results and may create snapshot copies of results stored on the edge computing device. The edge computing device may further cause the snapshot copies to be stored in an object-based data storage service of the service provider network. In some embodiments, snapshotting may not be performed, based on customer preferences.

[0053] FIG. 1 illustrates a quantum computing service of a service provider network that enables customers to access quantum computers that use multiple quantum computing technologies, wherein the quantum computing service includes a compilation module, according to some

embodiments.

[0054] Service provider network **100** includes quantum computing service **102**. In some embodiments, service provider network **100** may include data centers, routers, networking devices, etc., such as of a cloud computing provider network. In some embodiments, customers **104**, **106**, and **108** and/or additional customers of service provider network **100** and/or quantum computing service **102**, may be connected to the service provider network **100** in various ways, such as via a logically isolated connection over a public network, via a dedicated private physical connection, not accessible to the public, via a public Internet connection, etc.

[0055] Also, quantum computing service **102** is connected to quantum hardware providers **122**, **124**, **126**, and **128**. In some embodiments, quantum hardware providers **122**, **124**, **126**, and **128** may offer access to run quantum programs and/or quantum circuits on quantum computers that operate based on various different types of quantum computing technologies or paradigms, such as based on neutral atom, ion-trap, superconductive materials, photons, etc.

[0056] As discussed in more detail in FIG. **12**, in some embodiments, a service provider network **100** may be extended to include one or more edge computing devices physically located at quantum hardware provider locations, such as in a facility of quantum hardware providers **122**, **124**, **126**, and **128**. Physically locating an edge computing device of a service provider network **100** on premises at a quantum hardware provider facility may extend data security and encryption of the service provider network **100** into the quantum hardware providers **122**, **124**, **126**, and **128** facilities, thus ensuring the security of customer data. Also, physically locating an edge computing device of a service provider network **100** on premises at a quantum hardware provider facility may reduce latency between a compute instance of the service provider network and a quantum computer located at the quantum hardware provider facility. Thus, some applications, such as hybrid algorithms that are sensitive to network latencies may be performed by quantum computing service **102**, whereas other systems without co-located classical compute capacity at a hardware provider location may have too high of latencies to perform such hybrid algorithms efficiently.

[0057] In some embodiments, quantum computing service **102** includes one or more back-end API transport modules **110**. In some embodiments, a back-end API transport module **110** may be primarily implemented on edge computing devices of the quantum computing service that are located at the quantum hardware provider locations (such as edge computing devices **1204a**, **1204b**, **1204c**, and **1204d** illustrated in FIG. **12**). Also, in some embodiments, at least some of the back-end API transport functionality may be implemented on the one or more computing devices of the service provider network that implement the quantum computing service (such as computing devices in data center **1206a**, **1206b**, **1206c** illustrated in FIG. **12**).

[0058] Quantum programs and/or quantum circuits that have been translated by translation module **112** may be provided to back-end API transport module **110** in order for the translated quantum circuits to be transported to a quantum computer at a respective quantum hardware provider location. In some embodiments, the translated quantum programs and/or quantum circuits may further be compiled and sent as a compiled binary via the back-end API transport, or may be sent via the back-end API transport with accompanying compilation metadata indicating an ordered list of compilation passes to be used to compile the quantum programs and/or quantum circuits and indicating compilation parameters to be used in the respective compilation passes. For ease of explanation, the following discussion refers to quantum circuits, but the techniques described may equally be applied to quantum programs comprising one or more quantum circuits. Also, in some embodiments, a compilation module may further determine optimized compilation passes and/or compilation parameters for classical compute portions of quantum programs, wherein the quantum programs include a classical computing portion and quantum computing portion.

[0059] In some embodiments, back-end API transport **110** may be a non-public API that is accessible by an edge computing device of service provider network **100**, but that is not publicly available. In some embodiments, edge computing devices at the quantum hardware providers **122**,

124, **126**, and **128** may periodically ping a quantum computing service side interface to the back-end API transport **110** to determine if there are any quantum circuits (or batches of quantum circuits) waiting to be transported to the edge computing device. If so, the edge computing device may perform an API call to the back-end API transport **110** to cause the quantum circuit to be transported over a private connection to the edge computing device and scheduled for execution on a quantum computer. As discussed herein, in some embodiments, the quantum circuits may be transported with accompanying metadata indicating an optimized list of compilation passes and/or compilation parameters to be used to compile the quantum circuit for execution on a particular QPU. Also, the edge computing device may have been configured with a quantum machine image that enables the edge computing device to interface with a scheduling application of the quantum hardware provider, where the edge computing device is located, in order to schedule a time slot on the quantum computer of the quantum hardware provider to execute the quantum circuit via the back-end API transport **110**.

[0060] In some embodiments, a quantum computing service **102** may include, a quantum algorithm development kit **114** and/or user interface application programmatic interface (API) **114**, a translation module **112**, and a quantum compute simulator using classical hardware **118**.

Additionally, a data plane **107** of the quantum computing service **102** may include a results service **101** that receives execution performance results for previously compiled quantum circuits that were executed on particular QPUs of QHPs or executed on a particular local QPU. Additionally, the data plane **107** may include a QPU information service that receives information about QPUs at QHPs that are to be used to execute quantum circuits. For example, calibration information and/or other QPU specific information **111** may be provided from QHPs **122**, **124**, **126**, or **128**, via back-end API transport **110** to QPU information service **103**, wherein the QPU specific information **111** comprises information about particular QPUs of the QHPs. A control plane for the quantum computing service **102** may include a quantum development kit and/or user interface application programmatic interface (API) **114**, a translation module **112**, and a compilation module **105**. The compilation module **105** may use QPU information from the QPU information service **103** and execution results information from the results service **101** to determine optimized lists of compilation passes for compiling quantum circuits received or designed via quantum development kit and/or user interface API **114**. As mentioned above, in some embodiments, the compilation module may provide the optimized ordered list of compilation passes to a QHP for use in performing compilation at the QHP as shown via compilation pass list and or compilation pass parameters **113**. Also, in some embodiments, the compilation module may include a compiler and may use the optimized ordered list of compilation passes to locally compile a compiled binary for the quantum circuit and may provide the compiled binary to a QHP.

[0061] In some embodiments, results of executing the quantum circuit on the quantum computer at the quantum hardware provider location may be returned to the edge computing device at the quantum hardware provider location. The edge computing device and/or quantum computing service **102** may cause the results to be stored in a data storage system of the service provider network **100**. In some embodiments, results storage/results notification module **116** may coordinate storing results and may notify a customer, such as customer **104**, that the results are ready from the execution of the customer's quantum object, such as a quantum task, quantum algorithm, or quantum circuit. In some embodiments, results storage/results notification module **116** may cause storage space in a data storage service to be allocated to a customer to store the customer's results. Also, the results storage/results notification module **116** may specify access restrictions for viewing the customer's results in accordance with customer preferences.

[0062] In some embodiments, quantum compute simulator using classical hardware **118** of quantum computing service **102**, may be used to simulate a quantum algorithm or quantum circuit using classical hardware. For example, one or more virtual machines of a virtual computing service may be instantiated to process a quantum program or quantum circuit simulation job. In some

embodiments, quantum compute simulator using classical hardware **118** may fully manage compute instances that perform quantum circuit or quantum program simulation. For example, in some embodiments, a customer may submit a quantum circuit or quantum program to be simulated and quantum compute simulator using classical hardware **118** may determine resources needed to perform the simulation job, reserve the resources, configure the resources, etc. In some embodiments, quantum compute simulator using classical hardware **118** may include one or more “warm” simulators that are pre-configured simulators such that they are ready to perform a simulation job without a delay typically involved in reserving resources and configuring the resources to perform simulation.

[0063] In some embodiments, quantum computing service **102** includes quantum hardware provider recommendation/selection module **120**. In some embodiments, quantum hardware recommendation/selection module **120** may make a recommendation to a quantum computing service customer as to which type of quantum computer or which quantum hardware provider to use to execute a quantum object submitted by the customer. Additionally, or alternatively, the quantum hardware provider recommendation/selection module **120** may receive a customer selection of a quantum computer type and/or quantum hardware provider to use to execute the customer's quantum object, such as a quantum task, quantum algorithm, quantum circuit, etc. submitted by the customer or otherwise defined with customer input.

[0064] In some embodiments, a recommendation provided by quantum hardware provider recommendation/selection module **120** may be based on one or more characteristics of a quantum object submitted by a customer and one or more characteristics of the quantum hardware providers supported by the quantum computing service **102**, such as one or more of quantum hardware providers **122**, **124**, **126**, or **128**.

[0065] In some embodiments, quantum hardware provider recommendation/selection module may make a recommendation based on known data about previously executed quantum objects similar to the quantum object submitted by the customer. For example, quantum computing service **102** may store certain amounts of metadata about executed quantum objects and use such metadata to make recommendations. In some embodiments, a recommendation may include an estimated cost to perform the quantum computing task by each of the first and second quantum hardware providers. In some embodiments, a recommendation may include an estimated error rate for each of the first and second quantum hardware providers in regard to performing the quantum computing task. In some embodiments, a recommendation may include an estimated length of time to execute the quantum computing task for each of the first and second quantum hardware providers. In some embodiments, a recommendation may include various other types of information relating to one or more quantum hardware providers or any combination of the above.

[0066] In some embodiments, quantum compute simulator using classical hardware **118**, may allow a customer to simulate one or more particular quantum computing technology environments. For example, a customer may simulate a quantum circuit in an annealing quantum computing environment and an ion trap quantum computing environment to determine simulated error rates. The customer may then use this information to make a selection of a quantum hardware provider to use to execute the customer's quantum circuit.

[0067] FIG. 2A illustrates a more detailed view of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0068] Compilation module **105**, which may be the same or similar to compilation module **105** shown in FIG. 1, includes a quantum circuit graph generator **206**, that may generate a graphical representation of a quantum circuit, such as a directed acyclic graph of an input quantum circuit description **202**. The compilation module **105** also includes an optimization metric store **210** that stores information indicating what optimization metric(s) (M) are to be used to optimize compilation passes and/or compilation parameters for the quantum circuit **202**. In some embodiments, the quantum circuit description **202** may include quantum circuit information, such

as shown in FIG. 6, or the quantum circuit graph generator **206**, may generate a graph such as shown in FIG. 3 and additionally generate quantum circuit information as shown in FIG. 6. Note that in some embodiments, different optimization metrics (M) may be used for different customers and/or different quantum programs or quantum circuits submitted by a same customer. Additionally, compilation module **105** includes QPU information store **208** which stores QPU information received from a QHP via a back-end API transport, such as back-end API transport **110**. In some embodiments, the QPU information store **208** may also store QPU information for local QPUs of the quantum computing service **102**, such as local QPUs **1226** shown in FIG. 12. In some embodiment, the QPU information store **208** stores QPU information, such as shown in FIG. 5, for QPUs that are to be used to execute quantum circuits submitted by customers of quantum computing service **102**. Additionally, available compilation pass store **212**, stores available compilation passes, such as those shown in FIG. 4. Also, compilation constraints store **214** stores known constraints that must be enforced when compiling a given quantum program or a given quantum circuit, such as those shown in FIG. 7. Note, that the compilation passes shown in FIG. 4, the QPU information shown in FIG. 5, the quantum circuit information shown in FIG. 6, and the constraints shown in FIG. 7 are provided as example types of inputs that compilation optimization module **216** may use to determine an optimized ordered list of compilation passes and/or compilation parameters to be used to compile a particular quantum circuit for execution on a particular QPU. However, these examples are not intended to be exhaustive, such that other compilation passes, types QPU information, constraints, quantum circuit descriptions, etc. may be used by compilation optimization module **216** in order to determine an optimized ordered list of compilation passes and/or compilation parameters to be used to compile a particular quantum circuit for execution on a particular QPU.

[0069] The compilation optimization module **216** uses the quantum circuit graph from quantum circuit graph generator **206** and/or accompanying quantum circuit information along with QPU information from QPU information store **208**, and an optimization metric from optimization metric store **210** to determine an optimized list of compilation passes and/or compilation parameters to be used to compile quantum circuit **202**. The optimization may use any of the supported quantum passes stored in available compilation pass store **212** and may enforce the constraints included in compilation constraints store **214**. The output of the compilation optimization module **216** may be compilation pass list **220** and compilation parameters **222**, which may be provided to back-end API transport **110** for transport to a compiler of a QHP, or may be provided to local compiler **218** of quantum computing service **102**. In the latter case, a compiled binary **224** may be provided to the back-end API transport for transport to a QHP or may be provided to local quantum hardware device **130** of quantum computing service **102**. In some embodiments, the compilation pass list **220** and compilation parameters **222** may be provided to a simulator, such as quantum compute simulator using classical hardware **118**, for use in simulating execution of a particular quantum circuit on a particular type of QPU and/or a specific QPU.

[0070] In some embodiments, the quantum circuit description provided to the compilation optimization module **216** from quantum circuit graph generator **206** may be a directed acyclic graph (DAG) and an accompanying property set dictionary for the quantum circuit, such as quantum circuit property set dictionary **602** shown in FIG. 6. As shown in FIG. 6, the property set dictionary for the quantum circuit may indicate circuit statistics, such as layers, blocks (e.g., quantum circuit blocks of a quantum program), depth, and other information, such as an initial layout for the quantum circuit or quantum program.

[0071] In some embodiments, the QPU information may include a description of a particular quantum hardware device that is to be used to execute a particular quantum circuit. For example, FIG. 5 illustrates an example quantum hardware description **502**. In some embodiments, such a quantum hardware description may include: [0072] a list of physical qubits of the quantum hardware device; [0073] a quantum gate basis for a final output circuit to be implemented on the

quantum hardware device; [0074] a coupling map, such as a directed graph; [0075] a dictionary of gate error rates for gates implemented using the quantum hardware device; [0076] a dictionary of cross-talk error rates for cross-talk between nearby two-qubit gates of the quantum hardware device; or [0077] other relevant information describing performance of the quantum hardware device.

[0078] In some embodiments, available compiler (which may also be referred to in some situations as a transpiler) passes may include layout synthesis passes **402**, mapping compilation passes **404**, basis change passes **406** and optimization passes **408**. FIG. **4** lists examples of these types of compiler passes, though in some embodiments other passes may be supported. In some embodiments, two broad types of passes may be used: 1) transformational passes that make changes to the graphical representation of the quantum circuit being compiled (e.g., the DAG); and 2) analysis passes that make changes to the property set dictionary for the quantum circuit being compiled.

[0079] In some embodiments, constraints may include constraints that require at least some compilation passes to be performed in order to generate a compiled version of the quantum circuit that is executable on the particular QPU and other constraints that affect an order in which compiler passes are performed. For example, some compiler passes may require other compiler passes to have already been performed before the compiler passes can be performed. The first type of constraint may be referred to as an existence constraint, for example in order for a quantum circuit to be executed on a particular QPU the gates may have to be unrolled into gates that are physically executable on the particular QPU (e.g., higher level gates that do not correspond to physical gates of the QPU must be replaced with the physical gates actually supported by the QPU). Thus, in order for a quantum circuit to be executed on the particular QPU, an unroll higher level gates pass may be required to have been performed.

[0080] The determined compilation pass list **220** may satisfy the constraints for the particular QPU and include an optimized list of compilation passes. For example, the ordered list of compilation passes included in pass list **220** may ensure that a compiled binary for the quantum circuit is unrolled to a given basis of the QPU, mapped such that input hardware is mapped via a coupling map, and that the compilation passes are optimized with respect to optimization metric (M).

[0081] In some embodiments, compilation optimization module **216** may use a tabular learning approach, which comprises a search algorithm for problems in which the state-action space is small enough to fit in a table. The search may use flags to ensure existence and dependency constraints are satisfied. Example flags that may be associated with the quantum circuit during or subsequent to particular ones of the compilation passes may include: [0082] A multi-qubit unrolling flag. The multi-qubit unrolling flag may indicate whether or not multi-qubit gates have been un-rolled into lower-level gates in a given compilation pass. [0083] A custom definition unrolling flag. The custom definition unrolling flag may indicate whether or not custom definitions have been unrolled in a given compilation pass. [0084] A layout selection flag. The layout selection flag may be formatted as a tuple (e.g. [flag.sub.layout selection, flag.sub.physical qubit, D.sub.layout}). If the flag.sub.layout selection is set to equal 3, it means that all qubits are fully mapped and ancilla is fully mapped. If the flag.sub.layout selection is set to equal 2, it means qubits fully mapped. If the flag.sub.layout selection is set to equal 1, it means that qubits are partially mapped. If the flag.sub.layout selection is set to equal 0, it means unmapped. If the flag.sub.physical qubit flag is set, it indicates that all qubit names in the circuit (at the end of a given compilation pass with which the flag is associated) are physical qubit names. If the D.sub.layout flag is set, it means that the layout dictionary is updated for the currently mapped qubits. [0085] A routing flag. The routing flag also uses a tuple (e.g. [flag.sub.routing, D.sub.routing}). If the flag.sub.routing is set to equal 2, it means all two-qubit gates are fully mapped. If the flag.sub.routing is set to equal 1, it means that two-qubit gates are partially mapped. If the flag.sub.routing is set to equal 0, it means the two-qubit gates are unmapped. D.sub.routing is the current layer number before which all two-qubit

gates are mapped. [0086] Unrolling flag. The unrolling flag indicates whether or not the gates have been unrolled to the given basis of the QPU.

[0087] In some embodiments, as part of performing compilation, the flags are updated and/or verified via analysis compilation passes. For example, a “check map” analysis pass may verify that the routing flag is set to equal 1. In some embodiments, a general state of the quantum circuit subsequent to each compilation pass may be defined using the graphical representation of the quantum circuit (which may be modified by a given compilation pass), the dictionary properties of the quantum circuit (which may be modified by a given compilation pass), the current pass number (in the ordered list of compilation passes), and the state of the above flags for the quantum circuit (e.g. state of the multi-qubit unrolling flag, state of the custom definition unrolling flag, state of the layout selection flag, state of the routing flag, state of the unrolling flag). In some embodiments, the flags may be used to ensure a valid output state. For example, flags corresponding to existence constraints may be verified to be set before determining an optimized list of compilation passes is a valid list that will result in a valid compiled quantum circuit.

[0088] FIG. 2B illustrates a more detailed view of an example compilation optimization module of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0089] In some embodiments, compilation optimization module **216** includes a quantum circuit classification matching module **226** that associates a received quantum circuit with one of a plurality of pre-defined quantum circuit classifications. Each quantum circuit classification may have an associated ordered compilation pass list and/or compilation parameters, which may be stored in store **228**. The compilation optimization module **216** may return the appropriate ordered compilation pass list and/or compilation parameters corresponding to the quantum circuit classification determined by the quantum circuit classification matching module **226** for a given quantum circuit. In some embodiments, compilation optimization module **216** further includes classification and compilation pass list machine learning module **230**, which may learn new classifications and associated compilation pass lists and associated compilation parameters. Additionally, classification and compilation pass list machine learning module **230** may update existing classifications and associated compilation pass lists and associated compilation parameters.

[0090] FIG. 2C illustrates a more detailed view of another example compilation optimization module of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0091] In some embodiments, a compilation module **105** as shown in FIG. 2B may additionally or alternatively include a compilation optimization module **234** that includes an individualized quantum circuit search for compilation pass list and/or compilation pass parameters using machine learning module. Instead of employing a search for a general quantum circuit classification as described for compilation optimization module **216**, the machine learning algorithm may perform a search for the individual quantum circuit that is to be compiled.

[0092] FIG. 2D illustrates a more detailed view of yet another an example compilation optimization module of a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments.

[0093] In some embodiments, a compilation module **105** as shown in FIG. 2B or 2C may additionally, or alternatively, include a compilation optimization module **238** that includes a quantum circuit structure encoding module **240**, which determines a graphical representation of the quantum circuit, such as the directed acyclic graph shown in FIG. 3. Also, compilation optimization module **238** includes a store of compilation passes and/or compilation parameters that have been used for quantum circuits having particular quantum circuit structures as represented in a graphical representation. Additionally, compilation optimization module **238** includes a neural network that is configured to generate compilation pass lists and/or compilation parameters based

on learned relationships for the quantum circuit structures stored in **242** and a given quantum circuit structure as represented by a graphical representation of a quantum circuit generated by quantum circuit structure encoding **240**.

[0094] As discussed above, FIG. **3** illustrates an example directed acyclic graph that may be used to represent a quantum circuit, according to some embodiments. For example, a quantum circuit received from a customer may be formatted as a text definition, a circuit diagram, or may be formatted in various other formats. The received quantum circuit may be re-formatted into a directed acyclic graph, such as shown in FIG. **3**. For example, an input quantum circuit comprising an input state **302**, measurement step **304**, and gates **306**, **308**, and **310**, may be represented as a directed acyclic graph (DAG) as shown in FIG. **3**.

[0095] FIG. **8** illustrates an example process that may be followed by a compilation module of a quantum computing service to determine an optimized list of quantum circuit compilation passes to be used to compile a given quantum circuit, according to some embodiments.

[0096] At block **802**, a quantum computing service receives a quantum circuit to be compiled. At block **804**, the quantum computing service receives or obtains QPU information for QPUs that are to be used to execute quantum circuits, such as the quantum circuit received at **802**. At block **806**, a machine learning algorithm used by the quantum computing service accesses stored compilation information for previously compiled quantum circuits, such as quantum pass lists and/or compilation parameters used for the previously compiled quantum circuits, along descriptive information for the previously compiled quantum circuits, such as DAGs for the previously compiled quantum circuits and/or property set dictionaries for the previously compiled quantum circuits, such as is included in quantum circuit information **602** of FIG. **6**. Also, the stored compilation information may include performance information for the previously compiled quantum circuits, such as run-times, complexity (number of qubits and/or gates used to implement the previously compiled quantum circuits), error rates, etc. The data accessed at block **806** may function as training data for training a machine learning algorithm **230**, **236**, or a neural network **244** as described in FIGS. **2B-2D**.

[0097] In some embodiments, at block **808**, a compilation module of the quantum computing service generates an analysis friendly representation of quantum circuit (e.g., directed acyclic graph as shown in FIG. **3** and/or circuit information as shown in FIG. **6**). At block **810**, a compilation optimization module of the compilation module utilizes relationships determined via a machine learning algorithm to relate the analysis friendly representation of the quantum circuit (or portions thereof) to information about previously compiled quantum circuits in order to determine a list of optimized compilation passes and/or optimized compilation parameters. At block **812**, the compilation module provides an ordered list of compilation passes to be used to compile a compiled binary executable on a particular QPU for the quantum circuit. The compilation module may also provide a set of compilation parameters to be used in the compilation passes.

[0098] At blocks **814**, **816**, **818**, and **820** the provided list of compilation passes and/or compilation parameters is used to compile a compiled binary for the quantum circuit and/or simulate the quantum circuit. The provided list of compilation passes and/or compilation parameters may be used by a QHP compiler or a quantum computing service compiler. Also, the compiled binary may be used at a QPU of a QHP or a local QPU of the quantum computing service.

[0099] FIG. **9** illustrates an example compilation optimization process that may be used by a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments. At block **902**, a compilation optimization module classifies a quantum circuit to be compiled as belonging to a particular quantum circuit classification of a plurality of stored quantum circuit classifications. At block **904**, the compilation optimization module selects a list of compilation passes and/or compilation parameters to be used to compile the quantum circuit based on the particular classification determined for the quantum circuit to be compiled.

[0100] FIG. **10** illustrates another example compilation optimization process that may be used by a

quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments. At block **1002**, a compilation optimization module translates a quantum circuit into graphical representation (e.g., directed acyclic graph). At block **1004**, the compilation optimization module identifies commonalities between the graphical representation of the quantum circuit to be compiled and information for previously compiled quantum circuits. Then, at block **1006**, the compilation optimization module determines compilation passes and/or compilation parameters to be used to compile the quantum circuit based on the identified commonalities with previously compiled quantum circuits and stored information indicating compilation passes and/or compilation parameters used to compile the previously compiled quantum circuits.

[0101] FIG. **11** illustrates yet another example compilation optimization process that may be used by a quantum circuit compilation pass and/or compilation parameters determination module, according to some embodiments. At block **1102** the compilation optimization module performs an individualized search for an optimized list of compilation passes and/or compilation parameters for the quantum circuit to be compiled using a machine learning algorithm.

[0102] In some embodiments a given compilation module, such as compilation module **105**, may include multiple types of compilation optimization modules, such as compilation optimization modules **216**, **234**, and **238**, that implement processes as shown in FIGS. **9-11**. In some embodiments, different compilation optimization techniques may be used based on different customer requirements for compilation optimization and/or quantum circuit complexity. For example, some customers may require individually optimized compilation as described in FIG. **11**, while other customers may desire category-based optimization as described in FIG. **9**. In some embodiments, different levels of optimization may require different amounts of time to perform and/or different computing resources to perform. Therefore, some customers may opt-into using a particular type of optimization while other customers may opt-into using a different type of optimization. Though, in some embodiments, the quantum computing service may support multiple optimization types from which a customer may choose.

[0103] FIG. **12** illustrates edge computing devices of a quantum computing service physically located at quantum hardware provider locations, according to some embodiments.

[0104] In some embodiments, service provider network **100**, as illustrated in FIG. **1**, may include one or more data centers connected to each other via private or public network connections. Also, edge computing devices located at quantum hardware provider locations may be connected to a service provider network via private or public network connections. For example, service provider network **100** illustrated in FIG. **12** includes data centers **1206a**, **1206b**, and **1206c** that are connected to one another via private physical network links of the service provider network **100**. In some embodiments, a customer of the service provider network may also be connected via a private physical network link that is not available to the public to carry network traffic, such as a physical connection at a router co-location facility. For example, customer **1210** is connected to a router associated with data center **1206c** via direct connection **1224**. In a similar manner, edge computing devices located at quantum hardware provider locations may be connected to a service provider network via a private physical network link that is not available to carry public network traffic.

[0105] For example, edge computing device **1204a** located at quantum hardware provider location **1202a** is connected to a router at data center **1206a** via direct connection **1218**. In a similar manner, edge computing device **1204b** at quantum hardware provider location **1202b** is connected to a router at data center **1206b** via direct connection **1220**. Also, edge computing device **1204c** at quantum hardware provider **1202c** is connected to a router at data center **1206c** via direct connection **1222**.

[0106] Also, in some embodiments an edge computing device of a service provider network located at a quantum hardware provider location may be connected to the service provider network via a logically isolated network connection over a shared network connection, such as via the Internet or another public network. For example, edge computing device **1204d** at quantum hardware provider

location **1202d** is connected to data center **1206c** via a logically isolated network connection via network **1216**. In a similar manner, in some embodiments a customer, such as customer **1214**, may be connected to service provider network **100** via public network **1212**.

[0107] In some embodiments, a quantum computing service such as quantum computing service **102**, may be implemented using one or more computing devices in any of data centers **1206a**, **1206b**, **1206c**, etc. Also, the quantum computing service **102**, may provide customers, such as customer **1214** or customer **1210**, access to quantum computers in any of quantum hardware provider locations **1202a**, **1202b**, **1202c**, **1202d**, etc. For example, a customer may not be restricted to using a quantum hardware provider in a local region where the customer is located. Instead, the customer may be allocated compute instances instantiated on a local edge computing device located at a selected quantum hardware provider location, such that the location of the customer does not restrict the customer's access to various types of quantum computing technology-based quantum computers.

[0108] In some embodiments, one or more of the data centers **1206** may also include local quantum hardware devices, such as local QPUs **1226**.

Example Edge Computing Device Located at a Quantum Hardware Provider Location

[0109] FIG. **13** illustrates an example edge computing device connected to a quantum computing service, according to some embodiments.

[0110] Service provider network **100** and quantum computing service **102** may be similar to the service provider networks and quantum computing services described herein, such as in FIG. **1**. Also, edge computing device **1352** may be a similar edge computing device as any of the edge computing devices described previously, such as in FIG. **1** or **12**. Edge computing device **1352** is connected to service provider network **100** via network connection **1300**, which may be a logically isolated network connection via a public network, a dedicated physical non-public network link, or other suitable network connection.

[0111] Edge computing device **1352** includes network manager **1358**, storage manager **1360**, and virtual machine control plane **1356**.

[0112] In some embodiments, a back-end application programmatic interface (API) transport of an edge computing device, such as back-end API transport **1354** of edge computing device **1352** may ping a quantum computing service to determine if there are quantum circuits (e.g., quantum tasks) waiting to be transported to the edge computing device. The edge computing device may further use a non-public back-end API transport, such as back-end API transport **1354** to bring the quantum circuit into the edge computing device **1352**.

[0113] Additionally, for each customer, a back-end API transport of an edge computing device of a quantum computing service, such as back-end API transport **1354** of edge computing device **1352**, may cause a virtual machine to be instantiated to manage scheduling and results for a given quantum circuit pulled into the edge computing device from a back-end API. For example, virtual machine **1370** may act as an interface to the quantum hardware provider for a given customer of the quantum computing service. The edge computing device may be directly connected to a local non-public network at the quantum hardware provider location and may interface with a scheduling component of the quantum hardware provider to schedule availability (e.g., usage slots) on a quantum computer of the quantum hardware provider.

[0114] In some embodiments, the virtual machine **1370** may be booted with a particular quantum machine image that supports interfacing with the scheduling component of the quantum hardware provider, wherein different quantum hardware providers require different scheduling interfaces.

[0115] In some embodiments, virtual machine **1370** may be booted with a quantum circuit queuing component **1372**, a quantum circuit scheduling component **1376**, a component that manages a local storage bucket on the edge computing device to temporarily store results, such as temporary bucket **1374** and results manager **1378**. In some embodiments, quantum circuit scheduling component **1376** may order quantum circuits in quantum circuit queuing component in the order they are

received, wherein the received order enforces QoS guarantees by ordering the quantum tasks in the quantum task queue of the quantum computing service based on priorities determined using the QoS guarantees.

[0116] In some embodiments, an edge computing device, such as edge computing device **1352**, may support multi-tenancy. Also, in some embodiments, edge computing device **1352** may also instantiate virtual machines that execute classical computing tasks, such as a classical computing portion of a hybrid algorithm. For example, edge computing device **1352** also includes a virtual machine **1362** for another customer that performs classical compute portions of a hybrid algorithm and an additional virtual machine **1364** for the other customer which acts as a quantum hardware provider interface for the other customer.

[0117] In some embodiments, a back-end API transport of an edge computing device located a quantum hardware provider location may interface with a back-end API transport interface **110** of a computing device/router at a remote location where one or more computing devices that implement the quantum computing service are located.

[0118] Note that edge computing device **1352** may be physically located at quantum hardware provider premises **1350**, such as in a building of a quantum hardware provider facility.

[0119] In some embodiments, the components of virtual machine **1370** may be included in back-end API transport **1354**, and the back-end API transport **1354** may execute the related components within the back-end API transport without causing a separate VM **1370** to be instantiated.

[0120] In some embodiments, edge computing device **1352** may interface with QHP services **1308** which may include a results service **1310**, an invocation service **1312**, and a calibration data service **1314**. Also, the QHP services **1308** may interface with a quantum machine **1302**, which may include QPU **1304** and control software **1306**. In some embodiments, calibration data service **1314** may obtain QPU information about QPU **1304** and provide it to edge computing device **1352**, which in turn provides the QPU information to the quantum computing service via back-end API transport **1354**, such that the QPU information is added to QPU information of QPU information service **103**. In a similar manner, results may be routed from results service **1310** of the QHP service to the results service **101** of the quantum computing service **102**. These results and QHP information may be used by compilation module **105** as described above.

Illustrative Computer System

[0121] FIG. **14** is a block diagram illustrating an example computing device that may be used in at least some embodiments.

[0122] FIG. **14** illustrates such a general-purpose computing device **1400** as may be used in any of the embodiments described herein. In the illustrated embodiment, computing device **1400** includes one or more processors **1410** coupled to a system memory **1420** (which may comprise both non-volatile and volatile memory modules) via an input/output (I/O) interface **1430**. Computing device **1400** further includes a network interface **1440** coupled to I/O interface **1430**.

[0123] In various embodiments, computing device **1400** may be a uniprocessor system including one processor **1410**, or a multiprocessor system including several processors **1410** (e.g., two, four, eight, or another suitable number). Processors **1410** may be any suitable processors capable of executing instructions. For example, in various embodiments, processors **1410** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors **1410** may commonly, but not necessarily, implement the same ISA. In some implementations, graphics processing units (GPUs) may be used instead of, or in addition to, conventional processors.

[0124] System memory **1420** may be configured to store instructions and data accessible by processor(s) **1410**. In at least some embodiments, the system memory **1420** may comprise both volatile and non-volatile portions; in other embodiments, only volatile memory may be used. In various embodiments, the volatile portion of system memory **1420** may be implemented using any

suitable memory technology, such as static random-access memory (SRAM), synchronous dynamic RAM or any other type of memory. For the non-volatile portion of system memory (which may comprise one or more NVDIMMs, for example), in some embodiments flash-based memory devices, including NAND-flash devices, may be used. In at least some embodiments, the non-volatile portion of the system memory may include a power source, such as a supercapacitor or other power storage device (e.g., a battery). In various embodiments, memristor based resistive random access memory (ReRAM), three-dimensional NAND technologies, Ferroelectric RAM, magnetoresistive RAM (MRAM), or any of various types of phase change memory (PCM) may be used at least for the non-volatile portion of system memory. In the illustrated embodiment, program instructions and data implementing one or more desired functions, such as those methods, techniques, and data described above, are shown stored within system memory **1420** as code **1425** and data **1426**.

[0125] In some embodiments, I/O interface **1430** may be configured to coordinate I/O traffic between processor **1410**, system memory **1420**, and any peripheral devices in the device, including network interface **1440** or other peripheral interfaces such as various types of persistent and/or volatile storage devices. In some embodiments, I/O interface **1430** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **1420**) into a format suitable for use by another component (e.g., processor **1410**). In some embodiments, I/O interface **1430** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **1430** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **1430**, such as an interface to system memory **1420**, may be incorporated directly into processor **1410**.

[0126] Network interface **1440** may be configured to allow data to be exchanged between computing device **1400** and other devices **1460** attached to a network or networks **1450**, such as other computer systems or devices as illustrated in FIG. 1 through FIG. 13, for example. In various embodiments, network interface **1340** may support communication via any suitable wired or wireless general data networks, such as types of Ethernet network, for example. Additionally, network interface **1340** may support communication via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks, via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

[0127] In some embodiments, system memory **1320** may represent one embodiment of a computer-accessible medium configured to store at least a subset of program instructions and data used for implementing the methods and apparatus discussed in the context of FIG. 1 through FIG. 13. However, in other embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media. Generally speaking, a computer-accessible medium may include non-transitory storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD coupled to computing device **1400** via I/O interface **1430**. A non-transitory computer-accessible storage medium may also include any volatile or non-volatile media such as RAM (e.g., SDRAM, DDR SDRAM, RDRAM, SRAM, etc.), ROM, etc., that may be included in some embodiments of computing device **1400** as system memory **1420** or another type of memory. In some embodiments, a plurality of non-transitory computer-readable storage media may collectively store program instructions that when executed on or across one or more processors implement at least a subset of the methods and techniques described above. A computer-accessible medium may further include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **1440**. Portions or all of multiple computing devices such as that illustrated in FIG. 14 may be used to implement the

described functionality in various embodiments; for example, software components running on a variety of different devices and servers may collaborate to provide the functionality. In some embodiments, portions of the described functionality may be implemented using storage devices, network devices, or special-purpose computer systems, in addition to or instead of being implemented using general-purpose computer systems. The term “computing device”, as used herein, refers to at least all these types of devices, and is not limited to these types of devices.

CONCLUSION

[0128] Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g. SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc., as well as transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

[0129] The various methods as illustrated in the Figures and described herein represent exemplary embodiments of methods. The methods may be implemented in software, hardware, or a combination thereof. The order of method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

[0130] Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. It is intended to embrace all such modifications and changes and, accordingly, the above description to be regarded in an illustrative rather than a restrictive sense.

Claims

1.-20. (canceled)

21. A system, comprising: one or more computing devices configured to implement a quantum computing service, wherein the quantum computing service comprises: a quantum algorithm development kit user interface or an application programmatic interface (API); and a compilation module; wherein: the quantum algorithm development kit user interface or the application programmatic interface (API) is configured to: receive, from a customer of the quantum computing service, a quantum program comprising one or more quantum circuits to be executed via the quantum computing service; and the compilation module is configured to: obtain information for generating a list of compilation passes, wherein the information comprise at least one of: a quantum circuit description for a given one of the one or more quantum circuits; or information about a quantum processing unit (QPU) to be used to execute the given quantum circuit; and generate, based on the obtained information, the list of compilation passes, wherein the determined list of compilation passes comprise at least one layout synthesis pass; and the one or more computing devices that implement the quantum computing service are further configured to: cause the compilation passes of the determined list to be performed to compile one or more executable objects for the quantum program; and cause the quantum program to be executed using the one or more executable objects resulting from the performance of the compilation passes of the determined list.

22. The system of claim 21, wherein the generated list of compilation passes comprises one or more additional passes, wherein the one or more additional passes comprise at least one of: a mapping pass; a basis change pass; or an optimization pass.

23. The system of claim 21, wherein to generate the list of compilation passes, the one or more computing devices that implement the compilation module are configured to select a given layout synthesis pass from a set of layout synthesis passes.

24. The system of claim 21, wherein to generate the list of compilation passes, the one or more

computing devices that implement the compilation module are configured to: select a mapping pass from a set of mapping passes; select a basis change pass from a set of basis change passes; and select an optimization pass from a set of optimization passes.

25. The system of claim 21, wherein the one or more computing devices configured to implement the compilation module are further configured to: determine a set of compilation parameters to be used to compile the one or more executable objects for the quantum program.

26. The system of claim 21, wherein the list of compilation passes, when used to compile the one or more executable objects for the quantum program, results in one or more executable objects that are: decomposed in a basis matching a basis of the QPU to be used to execute the quantum program; and mapped to quantum hardware components of the QPU to be used to execute the quantum program.

27. The system of claim 21, wherein the one or more computing devices configured to implement the compilation module are further configured to: translate the one or more quantum circuits included in the quantum program into one or more respective directed acyclic graphs; identify commonalities in the one or more directed acyclic graphs for the one or more quantum circuits included in the quantum program and other directed acyclic graphs for previously compiled quantum circuits; and determine the list of compilation passes to be used for the one or more quantum circuits of the quantum program based on the identified commonalities to the previously compiled quantum circuits and respective lists of compilation passes used to compile the previously compiled quantum circuits.

28. A method, comprising: receiving, from a customer of a quantum computing service, a quantum program comprising one or more quantum circuits to be executed via the quantum computing service; obtaining information for generating a list of compilation passes, wherein the information comprise at least one of: a quantum circuit description for a given one of the one or more quantum circuits; or information about a quantum processing unit (QPU) to be used to execute the given quantum circuit; and generating, based on the obtained information, the list of compilation passes, wherein the determined list of compilation passes comprise at least one layout synthesis pass; causing the compilation passes of the determined list to be performed to compile one or more executable objects for the quantum program; and causing the quantum program to be executed using the one or more executable objects resulting from the performance of the compilation passes of the determined list.

29. The method of claim 28, wherein the generated list of compilation passes comprise one or more additional passes, wherein the one or more additional passes comprise at least one of: a mapping pass; a basis change pass; or an optimization pass.

30. The method of claim 28, wherein to generate the list of compilation passes, the method further comprises: selecting a given layout synthesis pass from a set of layout synthesis passes.

31. The method of claim 28, wherein to generate the list of compilation passes, the method further comprises: selecting a mapping pass from a set of mapping passes; selecting a basis change pass from a set of basis change passes; and selecting an optimization pass from a set of optimization passes.

32. The method of claim 28, wherein the method further comprises: determining a set of compilation parameters to be used to compile the one or more executable objects for the quantum program.

33. The method of claim 28, wherein the list of compilation passes, when used to compile the one or more executable objects for the quantum program, results in one or more executable objects that are: decomposed in a basis matching a basis of the QPU to be used to execute the quantum program; and mapped to quantum hardware components of the QPU to be used to execute the quantum program.

34. One or more non-transitory, computer-readable media storing program instructions that, when executed on or across one or more processors, cause the one or more processors to: receive, from a

customer of a quantum computing service, a quantum program comprising one or more quantum circuits to be executed via the quantum computing service; obtain information for generating a list of compilation passes, wherein the information comprise at least one of: a quantum circuit description for a given one of the one or more quantum circuits; or information about a quantum processing unit (QPU) to be used to execute the given quantum circuit; and generate, based on the obtained information, the list of compilation passes, wherein the determined list of compilation passes comprise at least one layout synthesis pass; cause the compilation passes of the determined list to be performed to compile one or more executable objects for the quantum program; and cause the quantum program to be executed using the one or more executable objects resulting from the performance of the compilation passes of the determined list.

35. The one or more non-transitory, computer-readable media of claim 34, wherein the generated list of compilation passes comprise one or more additional passes, wherein the one or more additional passes comprise at least one of: a mapping pass; a basis change pass; or an optimization pass.

36. The one or more non-transitory, computer-readable media of claim 34, wherein to generate the list of compilation passes, the program instructions, when executed on or across the one or more processors, cause the one or more processors to: select a given layout synthesis pass from a set of layout synthesis passes.

37. The one or more non-transitory, computer-readable media of claim 34, wherein to generate the list of compilation passes, the program instructions, when executed on or across the one or more processors, cause the one or more processors to: select a mapping pass from a set of mapping passes; select a basis change pass from a set of basis change passes; or select an optimization pass from a set of optimization passes.

38. The one or more non-transitory, computer-readable media of claim 34, wherein the program instructions, when executed on or across the one or more processors, cause the one or more processors to: determine a set of compilation parameters to be used to compile the one or more executable objects for the quantum program.

39. The one or more non-transitory, computer-readable media of claim 34, wherein the list of compilation passes, when used to compile the one or more executable objects for the quantum program, results in one or more executable objects that are: decomposed in a basis matching a basis of the QPU to be used to execute the quantum program; and mapped to quantum hardware components of the QPU to be used to execute the quantum program.

40. The one or more non-transitory, computer-readable media of claim 34, wherein to determine the list of compilation passes, the program instructions, when executed on or across the one or more processors, cause the one or more processors to: translate the one or more quantum circuits included in the quantum program into one or more respective directed acyclic graphs; identify commonalities in the one or more directed acyclic graphs for the one or more quantum circuits included in the quantum program and other directed acyclic graphs for previously compiled quantum circuits; and determine the list of compilation passes to be used for the one or more quantum circuits of the quantum program based on the identified commonalities to the previously compiled quantum circuits and respective lists of compilation passes used to compile the previously compiled quantum circuits.
